Stockholm University

Department of Meteorology

Numerical Methods in Atmospheric Sciences and Oceanography (MO8007)

Home Assignment 3

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Question

Exercise 1 - Non rotating 2-D SWE equations

Consider the non rotating 2-D SWE equations

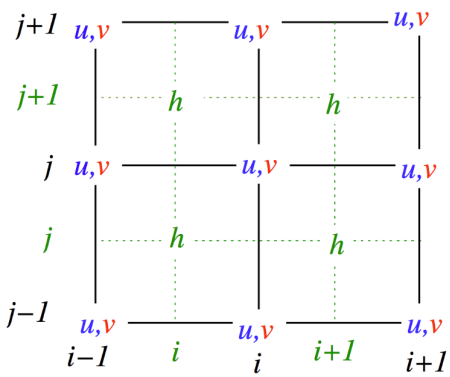

Figure 1: The Arakawa B grid.

(a) Discretise the equation on an Arakawa B-grid and apply the leapfrog scheme in time.

(b) Compute the stability criterion for this set up. For simplicity consider the case $\Delta x = \Delta y$.

1 The stability of the non-rotating 2D SWE on a B-grid

(a) The non-rotating 2-D SWE equations are given as

$$\begin{cases} \dfrac{\partial u}{\partial t} = -g\dfrac{\partial h}{\partial x} & \text{(1a)} \\ \dfrac{\partial v}{\partial t} = -g\dfrac{\partial h}{\partial y} & \text{(1b)} \\ \dfrac{\partial h}{\partial t} + H\left(\dfrac{\partial u}{\partial x} + \dfrac{\partial v}{\partial y}\right) = 0. & \text{(1c)} \end{cases}$$

Discretising the equations on an Arakawa B-grid by applying the leapfrog scheme one obtains

$$\begin{cases} \dfrac{u_{i,j}^{n+1} - u_{i,j}^{n-1}}{2\Delta t} = -g\dfrac{h_{i+1,j+1}^n + h_{i+1,j}^n - h_{i,j+1}^n - h_{i,j}^n}{2\Delta x} & \text{(2a)} \\ \dfrac{v_{i,j}^{n+1} - v_{i,j}^{n-1}}{2\Delta t} = -g\dfrac{h_{i,j+1}^n + h_{i+1,j+1}^n - h_{i+1,j}^n - h_{i,j}^n}{2\Delta y} & \text{(2b)} \\ \dfrac{h_{i,j}^{n+1} - h_{i,j}^{n-1}}{2\Delta t} = -H\left(\dfrac{u_{i,j}^n + u_{i,j-1}^n - u_{i-1,j}^n - u_{i-1,j-1}^n}{2\Delta x} + \dfrac{v_{i,j}^n + v_{i-1,j}^n - v_{i,j-1}^n - v_{i-1,j-1}^n}{2\Delta y}\right). & \text{(2c)} \end{cases}$$

(b) To solve for the stability criterion we apply a von Neumann stability analysis. We assume that the solution can be written as a Fourier mode of the form

$$\begin{pmatrix} u_{i,j}^n \\ v_{i,j}^n \\ h_{i,j}^n \end{pmatrix} = \begin{pmatrix} u_0 \\ v_0 \\ h_0 \end{pmatrix} e^{\mathfrak{I}(ki\Delta x + lj\Delta y - \omega n\Delta t)}, \tag{3}$$

where k and l are the wavenumbers in the x and y direction, respectively, ω is the frequency and $\mathfrak{I}$ is the imaginary unit.

One can simplify Equation 2 as

$$
\begin{aligned}
u_0 &= -\frac{1}{\lambda-\lambda^{-1}}\frac{gh_0\Delta t}{\Delta x}\left(e^{\mathfrak{I}k\Delta x}e^{\mathfrak{I}l\Delta y}+e^{\mathfrak{I}k\Delta x}-e^{\mathfrak{I}l\Delta y}-1\right)\\
v_0 &= -\frac{1}{\lambda-\lambda^{-1}}\frac{gh_0\Delta t}{\Delta y}\left(e^{\mathfrak{I}k\Delta x}e^{\mathfrak{I}l\Delta y}+e^{\mathfrak{I}l\Delta y}-e^{\mathfrak{I}k\Delta x}-1\right)\\
\left(\lambda-\lambda^{-1}\right)h_0 &= -\frac{Hu_0\Delta t}{\Delta x}\left(1+e^{-\mathfrak{I}l\Delta y}-e^{-\mathfrak{I}k\Delta x}-e^{-\mathfrak{I}k\Delta x}e^{-\mathfrak{I}l\Delta y}\right)\\
&\quad -\frac{Hv_0\Delta t}{\Delta y}\left(1+e^{-\mathfrak{I}k\Delta x}-e^{-\mathfrak{I}l\Delta y}-e^{-\mathfrak{I}k\Delta x}e^{-\mathfrak{I}l\Delta y}\right).
\end{aligned}
$$

One can simplify this by assuming $\Delta x = \Delta y$ and defining $\mu = \sqrt{gH}\,\Delta t/\Delta x$. Then inserting the expressions for u_0 and v_0 into the last equation gives

$$
\begin{aligned}
\left(\lambda-\lambda^{-1}\right)^2 &= \mu^2\left(e^{\mathfrak{I}(k+l)\Delta x}+e^{\mathfrak{I}k\Delta x}-e^{\mathfrak{I}l\Delta x}-1\right)\left(1+e^{-\mathfrak{I}l\Delta x}-e^{-\mathfrak{I}k\Delta x}-e^{-\mathfrak{I}(k+l)\Delta x}\right)\\
&\quad+\mu^2\left(e^{\mathfrak{I}(k+l)\Delta x}+e^{\mathfrak{I}l\Delta x}-e^{\mathfrak{I}k\Delta x}-1\right)\left(1+e^{-\mathfrak{I}k\Delta x}-e^{-\mathfrak{I}l\Delta x}-e^{-\mathfrak{I}(k+l)\Delta x}\right).
\end{aligned}
$$

Here one can make the neat observation by using the complex conjugates

$$
\begin{aligned}
(\lambda-\lambda^{-1})^2 &= \mu^2\left(e^{\mathfrak{I}k\Delta x}-1\right)(e^{\mathfrak{I}l\Delta x}+1)\left(1-e^{-\mathfrak{I}k\Delta x}\right)(1+e^{-\mathfrak{I}l\Delta x})\\
&\quad+\mu^2\left(e^{\mathfrak{I}l\Delta x}-1\right)(e^{\mathfrak{I}k\Delta x}+1)\left(1-e^{-\mathfrak{I}l\Delta x}\right)(1+e^{-\mathfrak{I}k\Delta x}).
\end{aligned}
$$

Now using the mathematical fact that

$$
\left(e^{\mathfrak{I}\theta}-1\right)\left(e^{-\mathfrak{I}\theta}-1\right)=4\sin^2\left(\frac{\theta}{2}\right)\quad\text{and}\quad\left(e^{\mathfrak{I}\theta}+1\right)\left(e^{-\mathfrak{I}\theta}+1\right)=4\cos^2\left(\frac{\theta}{2}\right), \tag{5}
$$

one can simplify the above expression to

$$
(\lambda-\lambda^{-1})^2=-16\mu^2\left[\sin^2\left(\frac{k\Delta x}{2}\right)\cos^2\left(\frac{l\Delta x}{2}\right)+\cos^2\left(\frac{k\Delta x}{2}\right)\sin^2\left(\frac{l\Delta x}{2}\right)\right].
$$

We now simply define the positive trigonometric expression as $\mathcal{A}^2$. Taking the square root of the above expression gives

$$
\lambda-\lambda^{-1}=\pm 4\mathcal{A}\mu\mathfrak{I}.
$$

Which gives the following expression for λ:

$$
\lambda=\pm 2\mathcal{A}\mu\mathfrak{I}\pm\sqrt{1-4\mathcal{A}^2\mu^2}.
$$

To evaluate the stability of the scheme we need to evaluate the magnitude of λ. We do this by considering if the square root is real or imaginary. If the square root is imaginary, then the magnitude of λ is given by

$$
|\lambda|^2=4\mathcal{A}^2\mu^2+\text{root}^2.
$$

This is greater than 1 for any value of μ which implies that the root must be real for the scheme to be stable. Thus if the root is real, then the magnitude of λ is given by

$$
|\lambda|^2=4\mathcal{A}^2\mu^2+\left(1-4\mathcal{A}^2\mu^2\right)=1.
$$

Thus the scheme is stable if the root is real, which implies that

$$
1-4\mathcal{A}^2\mu^2\geq 0\quad\Rightarrow\quad\mu\leq\frac{1}{2\mathcal{A}}.
$$

And since $\mathcal{A}^2$ is positive and has a maximum value of 1, we can thus conclude that the stability criterion.

Answer

The stability criterion for the non-rotating 2D SWE on a B-grid is given by $\mu \leq \frac{1}{2\mathcal{A}}$, where μ is the Courant number and $\mathcal{A}$ is a positive trigonometric expression defined as $\mathcal{A}^2=\sin^2\left(\frac{k\Delta x}{2}\right)\cos^2\left(\frac{l\Delta x}{2}\right)+\cos^2\left(\frac{k\Delta x}{2}\right)\sin^2\left(\frac{l\Delta x}{2}\right)$.

This implies that the scheme is stable if the Courant number $\mu \leq \frac{1}{2}$.